



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

08-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	D	B	A	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	D	A,C	A,B,C,D	B,C	A,B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	5.00	9.00	0.50	2.42	0.18

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	C	C	D	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	A,D	A,C	B	A,B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.41	12.24	4.00	5.00	4.00	1.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	C	B	A	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	A,C	A,C,D	A,B,C	A,B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.00	1.00	0.00	141.00	1.69	24.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A)**
 $q_1 = \frac{C_1}{C_1 + C_2} q$ where q is the total charge

$$\therefore \frac{q_1}{q} = \frac{1}{3} = \frac{C_1}{C_1 + C_2}$$

$$\therefore C_1 + C_2 = 3C_1$$

$$\text{Also } \frac{q_2}{q} = \frac{2}{3} = \frac{C_2}{C_1 + C_2}$$

$$\therefore 3C_2 = 2 \times 3C_1 (\because C_1 + C_2 = 3C_1)$$

$$\text{i.e., } C_2 = 2C_1$$

When the distance is altered between the coatings of C_1

$$\frac{q'_1}{q} = \frac{1}{4} = \frac{C'_1}{C'_1 + C_2}$$

$$\therefore C'_1 + C_2 = 4C'_1$$

$$\text{Also } \frac{3}{4} = \frac{C'_2}{C'_1 + C_2}$$

$$\text{Hence } 3 \times 4C'_1 = 4C_2$$

$$\text{i.e. } C_2 = 3C'_1$$

From (1) and (3) we get,

$$C'_1 = \frac{2}{3} C_1$$

$$= \frac{2}{3} \times 3 \times 10^{-3} = 2\text{mF}$$

2. **Ans (D)**

Let V be the volume of liquid flown out of container

$$V = \frac{AL}{3} [\gamma \Delta T + 2\gamma \Delta T + 3\gamma \Delta T]$$

$$- \left[A(1 + \gamma \Delta T)L \left(1 + \frac{\gamma}{2} \Delta T \right) - AL \right]$$

$$= 2AL\gamma \Delta T - AL \left(1 + \frac{3\gamma}{2} \Delta T \right) + AL = \frac{AL\gamma \Delta T}{2}$$

3. Ans (B)

Since magnetic field is a linear function of x So

$$B_{\text{avg}} = B = \frac{B_0 R}{2}$$

$$\Rightarrow R = \text{maximum horizontal distance} = \frac{\mu}{qB}$$

$$\Rightarrow R = \frac{\mu}{q \frac{B_0 R}{2}} \Rightarrow R = \sqrt{\frac{2\mu}{qB_0}}$$

Alternative solution :

$$\text{or } \vec{v} = v_x \hat{i} + v_y \hat{j}$$

$$\vec{B} = B_0 x \hat{k}$$

$$\vec{F} = -q(\vec{v} \times \vec{B})$$

$$\Rightarrow m v_x \frac{dv_x}{dx} = -q v_y B_0 x$$

$$m v_y \frac{dv_y}{dy} = -q v_x B_0 x$$

$$\text{Also, } v_y^2 = u^2 - v_x^2$$

$$\Rightarrow \int_0^x \frac{-q B_0 x dx}{m} = \int_u^0 \frac{v_x dv_x}{\sqrt{u^2 - v_x^2}}$$

$$\Rightarrow x_{\text{max}} = R = \sqrt{\frac{2\mu}{qB_0}}$$

4. Ans (A)

As process is isochoric, let temperature at any

instant of time t be T .

$$\frac{dQ}{dt} = \frac{KA \cdot (2T_0 - T)}{d} = \frac{5}{2} \frac{RdT}{dt}$$

$$\Rightarrow \int_{T_0}^T \frac{dT}{2T_0 - T} = \int_0^t \frac{2}{5} \frac{KA}{dR} dt$$

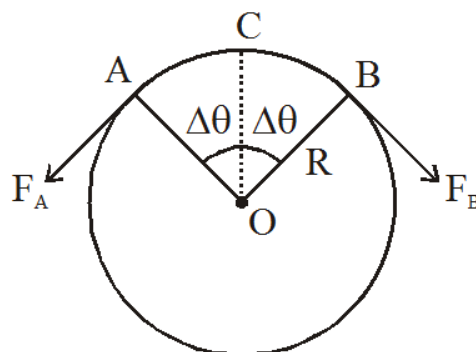
$$\Rightarrow T = T_0(2 - e^{-\frac{2KA t}{5dR}})$$

5. Ans (C)

To find tension in the rope, net force along

$$CO = F_A \sin \Delta\theta + F_B \sin \Delta\theta$$

(components perpendicular to CO cancel each other)



But $F_A = F_B = F$, tension in the string.

$$\therefore 2F \sin \Delta\theta = m \times AB \times R\omega^2,$$

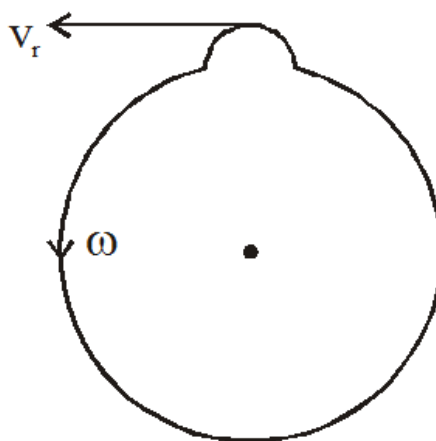
$AB = R 2\Delta\theta$ and m = linear density of the rope.

$$= 2m R_2 \omega^2 \Delta\theta$$

$$\text{For } \Delta\theta \ll 1, F = mR^2\omega^2$$

Now a kink is formed, moving in anti-clockwise direction relative to an observer in the ground speed of the kink relative to the

$$v_k = \sqrt{\frac{F}{m}} = R\omega$$



For an observer on the ground,

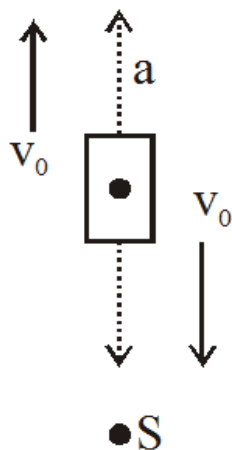
$$v_g = v_k + v_r = 60 \text{ ms}^{-1} \text{ (given)}$$

$$\therefore 2\omega R = 60 \text{ or } \omega R = 30 \text{ ms}^{-1}$$

$$\text{Hence tension } F = 0.2 \times 30 \times 30 = 180 \text{ N}$$

6. Ans (B)

When the microphone crosses the equilibrium position with maximum speed v_Q towards the source S, the frequency detected becomes maximum.



(i.e.,) $f_{\max} = f_0 \left[1 + \frac{v_0}{v} \right]$ v – speed of sound.

Similarly, when it crosses the equilibrium position away from the source, frequency detected becomes minimum.

(i.e.,) $f_{\min} = f_0 \left(1 + \frac{v_0}{v} \right)$

Given : $f_{\max} - f_{\min} = 2$

$$= f_0 \left(1 + \frac{v_0}{v} - 1 + \frac{v_0}{v} \right) = \frac{2f_0 v_0}{v}$$

$$\therefore v_0 = \frac{2v}{2f_0} = \frac{v}{f_0} = \frac{330}{660} = \frac{1}{2} \text{ms}^{-1}$$

Let a be the amplitude of SHM so that the angular velocity

$$\omega = \frac{v_0}{a} \text{ or } a = \frac{v_0}{\omega} = \frac{v_0 \times T}{2\pi}$$

$$a = \frac{2}{2} \times \frac{1}{2\pi} = \frac{0.5}{3.14} = \frac{50}{314}$$

$$= 0.16 \text{ m} = 16 \text{ cm}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (D)

According to the given problem,

$$x = a \sin \omega t \text{ and } \dots(i)$$

$$y = a \sin (2 \omega t + \phi) \dots(ii)$$

where ϕ is the initial phase difference to be determined.

The path followed by the particle as a result of superposition is

$$x^2 = -\frac{a}{2}(y - a)$$

Substituting for x and y from (i) and (ii), we get

$$a^2 \sin^2 \omega t = +\frac{a}{2}(a - y)$$

$$= \frac{a}{2} [a - a \sin(2\omega t + \phi)]$$

$$\therefore 2 \sin^2 \omega t = [1 - \sin(2\omega t + \phi)]$$

$$= [\sin^2 \omega t + \cos^2 \omega t - \sin(2\omega t + \phi)]$$

$$\text{i.e., } \sin^2 \omega t - \sin^2 \omega t = \sin(2\omega t + \phi)$$

$$\Rightarrow \cos^2 \omega t - \sin^2 \omega t = \sin(2\omega t + \phi)$$

$$\Rightarrow \cos 2\omega t = \sin(2\omega t + \phi)$$

$$\text{i.e., } \sin\left(2\omega t + \frac{\pi}{2}\right) = \sin(2\omega t + \phi)$$

$$\therefore \phi = \frac{\pi}{2}$$

8. Ans (A,C)

$$Ran^2, v\alpha \frac{1}{n} \text{ and } E\alpha \frac{1}{n^2}$$

9. Ans (A,B,C,D)

$$y = 2A \sin(kx - \omega t)$$

$$\left(\frac{\partial K}{\partial x}\right)_{\text{antinodes}} = 2\mu A^2 \omega^2 \sin^2 \omega t$$

$$\left(\frac{\partial U}{\partial x}\right)_{\text{nodes}} = 2\mu A^2 \omega^2 \cos^2 \omega t$$

$$\left(\frac{\partial K}{\partial x}\right)_{\text{antinodes}} + \left(\frac{\partial U}{\partial x}\right)_{\text{nodes}} = 2\mu A^2 \omega^2 = \text{constant}$$

For a point midway between a node and an adjacent antinode

$$\frac{\partial K}{\partial x} = \mu A^2 \omega^2 \sin^2 \omega t$$

$$\frac{\partial U}{\partial x} = \mu A^2 \omega^2 \cos^2 \omega t$$

$$\frac{\partial K}{\partial x} + \frac{\partial U}{\partial x} = \mu A^2 \omega^2 = \text{constant}$$

$$\text{Let } y = A \sin(kx - \omega t) + 2A \sin kx \cos \omega t$$

$$\frac{\partial K}{\partial x} = \frac{1}{2} \mu \left(\frac{\partial y}{\partial t}\right)^2 = \frac{1}{2} \mu$$

$$[-2A\omega \sin kx \sin \omega t - A\omega \cos(kx - \omega t)]^2$$

$$\frac{\partial U}{\partial x} = \frac{1}{2} T \left(\frac{\partial y}{\partial x}\right)^2 = \frac{1}{2} V^2 \mu$$

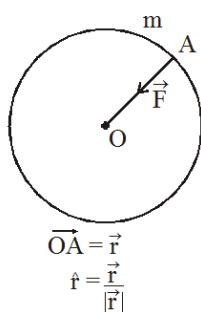
$$[-2Ak \cos kx \cos \omega t - Ak \cos(kx - \omega t)]^2$$

10. Ans (B,C)

Work done against the force in taking the particle from the orbit of radius r to infinity is

$$W = \int_r^\infty \vec{F} \cdot (d\vec{r}) = -K \int_r^\infty \frac{dr}{r^3} = +\frac{K}{2} \left[\frac{1}{r^2}\right]_r^\infty = -\frac{K}{2r^2}$$

$$\text{But } W = -[U_\infty - U_r] = U_r = -\frac{K}{2r^2}$$



$$\therefore \text{potential energy at } r, U_r = +W - \frac{K}{2r^2}$$

$$\text{Also, work done } W = K_f - K_i = 0 - K_i = -\frac{K}{2r^2}$$

$$\therefore \text{kinetic energy at the orbit} = \frac{K}{2r^2}$$

$$\text{Thus, total energy } E = K.E. + P.E. = 0$$

Now, if ω is the angular velocity, then,

$$\frac{1}{2} m r^2 \omega^2 = \frac{K}{2r^2} \text{ or } \omega^2 \propto \frac{1}{r^4} \text{ (i.e.,) } \omega \propto \frac{1}{r^2}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} \propto r^2$$

11. Ans (A,B,C)

(i) Path difference due to the glass slab,

$$\Rightarrow \Delta x = (\mu - 1)t = (1.5 - 1)t = 0.5t$$

Due to this slab, 5 red fringes have been shifted upwards. Therefore

$$\Delta x = 5\lambda_{\text{red}} \text{ or } 0.5t = (5)(7 \times 10^{-7} \text{ m})$$

$$\therefore t = \text{thickness of glass slab} = 7 \times 10^{-6} \text{ m}$$

(ii) Let μ' be the refractive index for green light then, $\Delta x' = (\mu' - 1)t$

Now the shifting is of 6 fringes of red light. Therefore,

(iii) In part (i), shifting of 5 bright fringes was equal to 10^{-3} m , which implies that

$$5\omega_{\text{red}} = 10^{-3} \text{ m} \quad [\text{Here } \omega = \text{Fringe width}]$$

$$\Rightarrow \omega_{\text{red}} = \frac{10^{-3}}{5} \text{ m} = 0.2 \times 10^{-3} \text{ m}$$

$$\text{Now since } \omega = \frac{\lambda D}{d} \text{ or } \omega \propto \lambda$$

$$\Rightarrow \frac{\omega_{\text{green}}}{\omega_{\text{red}}} = \frac{\lambda_{\text{green}}}{\lambda_{\text{red}}}$$

$$\therefore \omega_{\text{green}} = \omega_{\text{red}} \frac{\lambda_{\text{green}}}{\lambda_{\text{red}}} = (0.2 \times 10^{-3}) \left(\frac{5 \times 10^{-7}}{7 \times 10^{-7}}\right)$$

$$\Rightarrow \omega_{\text{green}} = 0.143 \times 10^{-3} \text{ m}$$

$$\therefore \Delta \omega = \omega_{\text{green}} - \omega_{\text{red}} = (0.143 - 0.12) \times 10^{-3} \text{ m}$$

$$\Rightarrow \Delta \omega = -5.71 \times 10^{-5} \text{ m}$$

$$7\mu, 1.6, 400/7\mu \text{ (decrease)}$$

12. Ans (A,C,D)

$$\vec{B} = \mu_0 n \ell$$

$$\therefore \vec{B} = \mu_0 n \alpha t$$

Now inside the solenoid

$$\vec{E} \cdot 2\pi r = \pi r^2 \cdot \mu_0 \alpha = |\vec{E}| = \frac{\mu_0 n \alpha r}{2}$$

$$\text{Outside the solenoid } \frac{d}{dt} (\beta \cdot \pi R^2) = E \cdot 2\pi r$$

$$= E = \frac{\pi R^2 \cdot \mu_0 n \alpha}{2\pi r} = \frac{\mu_0 n \alpha R^2}{2r}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (8.00)

$$W = U_f - U_i = \frac{Q_0^2}{2C} - \frac{Q_0^2}{2C},$$

$$v \propto \frac{1}{\sqrt{C}} \Rightarrow C' = \frac{C}{9}$$

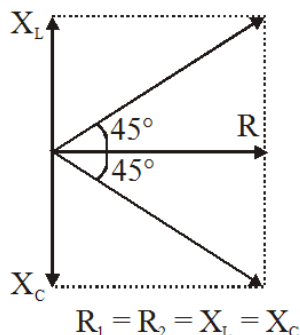
$$= \frac{9Q_0^2}{2C} - \frac{Q_0^2}{2C} = 9E - E = 8E$$

$$\therefore n = 8$$

2. **Ans (5.00)**

Peak current through R_1 , $I_1 = \frac{50}{10\sqrt{2}} = \frac{5}{\sqrt{2}} \text{ A}$

Peak current through R_2 , $I_2 = \frac{50}{10\sqrt{2}} = \frac{5}{\sqrt{2}} \text{ A}$



Phase difference between I_1 and I_2 is $\frac{\pi}{2}$

$\therefore$ Peak current through the source is

$$I = \sqrt{I_1^2 + I_2^2} = \sqrt{\left(\frac{5}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = \frac{5}{\sqrt{2}} \times \sqrt{2}$$

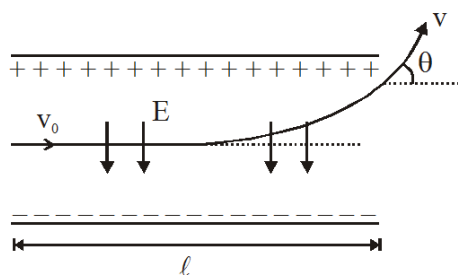
$\therefore I = 5 \text{ A}$

3. **Ans (9.00)**

Since, electron is accelerated through a potential difference V , its initial velocity v_0 is given by

$$\frac{1}{2}mv_0^2 = eV$$

or $v_0 = \sqrt{\frac{2eV}{m}} \dots(i)$



Since, initial velocity is parallel to plates or normal to the direction of electric field, component of velocity parallel to plates remains constant as v_0 .

Hence, time taken by the electron to cross electric field is $t_0 = \frac{l}{v_0}$

Now consider motion of electron, normal to plates.

At some instant t , its acceleration $= \frac{eE}{m} = \frac{eat}{m}$

Let velocity component normal to plates be v_y

$$\therefore \frac{d}{dt}v_y = \frac{eat}{m}$$

$$\therefore \int_0^{v_y} dv_y = \frac{ea}{m} \int_0^{t_0} t dt$$

or $v_0 = \frac{ea}{2m}t_0^2 = \frac{a\ell^2}{4V} \dots(ii)$

If θ is angular deviation of electron from its initial direction of motion, then pitch of its helical path.

$$p = \frac{2\pi m}{eB} v \cos(90 - \theta)$$

$$\therefore p = \frac{2\pi m}{eB} v \sin \theta = \frac{2\pi m}{eB} v_y$$

or $p = \frac{\pi m a \ell}{2eBV}$

$p = 9 \text{ mm}$

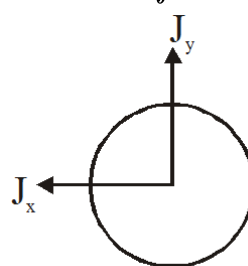
4. **Ans (0.50)**

Since collision is elastic

$$e = 1 = \frac{v_x}{v_0} \Rightarrow v_x = v_0$$

$$\therefore J_x = 2mv_0$$

Also $J_y = \mu \int N dt = \mu J_x = mv_y$



$$\therefore v_y = \frac{\mu \times 2mv_0}{m} = 2\mu v_0$$

Now for $R_{\max} \theta = 45^\circ$

$$\therefore v_y = v_x$$

$$\Rightarrow \mu = \frac{1}{2}$$

5. **Ans (2.42)**

Least count of screw gauge $= \frac{1}{100} \text{ mm} = 0.01 \text{ mm}$

Diameter of the wire $= (1 + 25 \times 0.01) \text{ mm} = 0.125 \text{ cm}$

Since $Y = \frac{4T\ell}{\pi d^2 \delta \ell}$

$$\therefore \frac{\Delta Y}{Y} = \frac{\Delta \ell}{\ell} + \frac{2\Delta d}{d} + \frac{\Delta(\delta \ell)}{\delta \ell}$$

$$= \frac{0.01}{50} + \frac{2 \times 0.001}{0.125} + \frac{0.001}{0.125} = 0.0242$$

Percentage error $= \frac{\Delta Y}{Y} \times 100 = 2.42$

6. **Ans (0.18)**

$$\vec{\tau}_f + \vec{\tau}_B = 0$$

$$0 = \vec{R} \times \vec{f}_f + \int \vec{r} \times (I d\vec{\ell} \times \vec{B})$$

$$= R f_r \sin \theta (+\hat{k}) + \int_0^R I B r dr (-\hat{k})$$

$$= R f_r (+\hat{k}) + \frac{I R^2 B}{2} (-\hat{k})$$

$$f_r = \frac{I R B}{2} = 0.18 \text{ N} = 18 \times 10^{-2} \text{ N}$$

PART-2 : CHEMISTRY

SECTION-I (i)

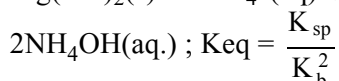
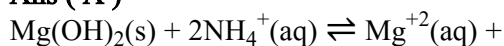
1. **Ans (C)**

$$\Delta H = qp = 3 \times (-35) = -105 \text{ kcal}$$

$$\Delta U = \Delta H = -\Delta n_g RT = -103.2 \text{ kcal}$$

$$w = U - q = -103.2 - (-105) = 1.8 \text{ kcal}$$

2. **Ans (A)**

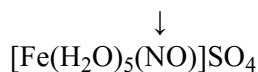
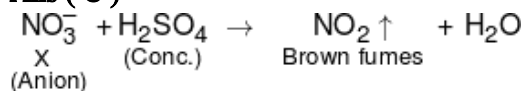


$$\frac{\left(\frac{0.15 \times 0.1}{0.5}\right) \left(\frac{0.35 \times 0.1}{0.5}\right)^2}{\left(\frac{n}{0.5}\right)^2} \leq \frac{1.2 \times 10^{-11}}{(2 \times 10^{-5})^2}$$

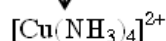
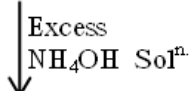
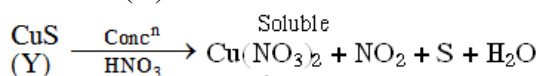
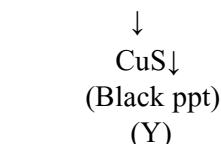
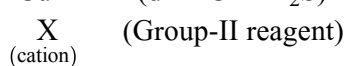
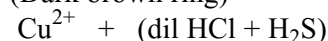
$$n \geq 0.035$$

$$= \frac{0.035}{2} \times 132 \text{ gm min. mass}$$

3. **Ans (C)**



(Dark brown ring)



Deep blue colour solution.



4. **Ans (C)**

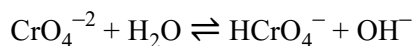
$[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_6]$ can interchange ligands so it can show coordination isomerism.

PART-2 : CHEMISTRY

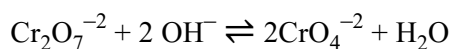
SECTION-I (ii)

7.

Ans (A,C,D)



$$K_{eq} = \frac{K_w}{K_1} = \frac{10^{-14}}{10^{-6.5}} = 3.16 \times 10^{-8}$$



$$K_{eq} = \frac{K_1^2}{K_2 K_w^2} = 4.34 \times 10^{13}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (1.41)**

$$\frac{25.7}{3.84} = \frac{4(64 \times 0.75 + 65 \times 0.25)}{6 \times 10^{23} \times a^3}$$

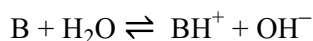
$$a^3 = 64 \times 10^{-24} \text{ cm}^3$$

$$a = 4 \times 10^{-8} \text{ cm} = 4 \text{ \AA}$$

$$\sqrt{2}a = 4r \Rightarrow r = \frac{\sqrt{2} \times 4}{4} \text{ \AA} = 1.41 \text{ \AA}$$

2. **Ans (12.24)**

$$\Lambda_m = \frac{1000 \times 1.53 \times 10^{-4}}{0.0306} = 5 \text{ Scm}^2 \text{ mol}^{-1}$$



$$\Lambda_m^\infty = 250$$

$$\alpha = \frac{5}{250} = \frac{1}{50}$$

$$K_b = C\alpha^2 = 0.0306 \times \left(\frac{1}{50}\right)^2 = 1.224 \times 10^{-5}$$

$$= 12.24 \times 10^{-6}$$

6. **Ans (1.00)**

Only statement

1. is correct.

2. FeSiO_3 is main slag

3. Roasting is done to convert FeO to Fe_2O_3 which does not form slag and hence is not wasted.

4. $\text{CuSO}_4 + \text{SO}_2$

7. Cu^{2+} is oxidising agent.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

Coefficient of $x^{60} = (-6) + (-1)(-5) + (-2)(-4) + (-1)(-2)(-3) = 1$

2. **Ans (C)**

$y = 2x$; $x^2 - xy + 2y^2 = 28$

solving the point of intersection are (2, 4) and (-2, -4)

For 1st curve, $2 = m_1$ (1)

for 2nd curve, $0 = m_2$ (2)

$\therefore \tan\theta = 2$

3. **Ans (B)**

$$\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} + \frac{1}{d+\omega} = \frac{1}{\omega} \quad \dots(i)$$

Taking conjugate we get,

$$\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} + \frac{1}{d+\omega^2} = \frac{1}{\omega^2} \quad \dots(ii)$$

Taking difference,

$$\sum \left(\frac{1}{a+\omega} - \frac{1}{a+\omega^2} \right) = \frac{1}{\omega} - \frac{1}{\omega^2}$$

$$\Rightarrow (\omega^2 - \omega) \sum \frac{1}{(a+\omega)(a+\omega^2)} = \omega^2 - \omega$$

$$\Rightarrow \sum \frac{1}{a^2 - a + 1} = 1$$

4. **Ans (A)**

If $f: [a, b] \rightarrow [c, d]$

$$\text{then, } \int_a^b f(x) dx + \int_c^d f^{-1}(x) dx = bd - ac$$

$$\therefore \int_0^1 f(x) dx + \int_1^3 f^{-1}(x) dx = 3$$

5. **Ans (C)**

Let point of intersection of tangents be (h, k)

$\therefore$ Equation of chord of contact will be : $xh + 2yk = 6$... (1)

and equation of tangent at $\frac{x^2}{4} + y^2 = 1$ will be :

$$\frac{x}{2} \cos\theta + y \sin\theta = 1 \quad \dots(2)$$

$$\therefore h = 3\cos\theta \text{ and } k = 3\sin\theta$$

$$\therefore h^2 + k^2 = 9$$

6. **Ans (B)**

For each element of set A, there are 4 possibilities i.e.,

(i) element of B but not element of C

(ii) element of C but not element of B

(iii) neither element of B nor element of C

(iv) element of B and C

Total = 4^5 and if $B \cap C$ is empty then ways = 3^5

favourable = $4^5 - 3^5$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A,C,D)**

$$f(x) = x^3 - 3ax^2 + 3(5a-6)x + 2$$

$$f'(x) = 3(x^2 - 2ax + (5a-6))$$

$$\text{Let } g(x) = x^2 - 2ax + 5a - 6$$

(A) $g(x) = 0$ has real and positive roots.

$$D > 0 \text{ and } a > 0 \text{ and } 5a - 6 > 0$$

$$a^2 - 5a + 6 > 0 \cap a > 0 \cap a > 6/5$$

$$\Rightarrow a \in (6/5, 2) \cup (3, \infty)$$

(B) $g(x) = 0$ has real and negative roots

$$D > 0 \text{ and } a < 0 \text{ and } a > 6/5$$

not possible

(C) $g(x) = 0$ has one root 0 and other is positive

$$a = 6/5 \text{ and other root is } 12/5$$

(D) $g(x) = 0$ has one root 0 and other is negative.

not possible.

8. Ans (A,C)

Let 7 number as $x_1, x_2, x_3 \dots x_7$ and these number can be chosen from 1 to 100 and $|x_i - x_j| \geq 7$

$$\begin{array}{ccccccc} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ d_8 & d_1 & d_2 & d_3 & d_4 & d_5 & d_6 & d_7 \end{array}$$

gap between any two number be $d_1, d_2 \dots d_6$

$$d_1 + d_2 + d_3 + d_4 + d_5 + d_6 + d_7 + d_8 = 93$$

where $d_1 \geq 7, d_2 \geq 7, d_3 \geq 7, d_4 \geq 7, d_5 \geq 7, d_6 \geq 7,$

$$d_7 \geq 0, d_8 \geq 0$$

so $d_i = t_i + 7, t_i \geq 0, i \in \{1, 2, 3, 4, 5, 6\}$

$$t_1 + t_2 + t_3 + t_4 + t_5 + t_6 + t_7 + t_8 = 93 - 7 \times 6 = 51$$

number of non negative integral solution

$$= {}^{51+8-1}C_7 = {}^{58}C_7$$

$$n = 58$$

9. Ans (A,C,D)

4 defective + 11 correct

P(getting exactly three defective in 8 record)

$$= \frac{{}^4C_3 \cdot {}^{11}C_5}{{}^{15}C_8}$$

P(9^{th} examined is last defective) =

$$\frac{{}^4C_3 \cdot {}^{11}C_5}{{}^{15}C_8} \times \frac{1}{7}$$

10. Ans (A,B,C)

Let the normal be $y = mx - 4m - 2m^3$

$$\Rightarrow 0 = 6m - 4m - 2m^3 \Rightarrow m = 0, 1, -1$$

$$A(0, 0); B(2, 4); C(2, -4)$$

$$\text{Area} = 8$$

$$\text{Centroid} \equiv (4/3, 0), \quad \text{circumcentre} \equiv (5, 0).$$

11. Ans (A,B,C)

$$|3\sin x - 4\cos x| = t$$

$$t^2 - (b^2 + b + 5)t + (b + 3)(b^2 + 2) = 0$$

$$\Rightarrow t^2 - (b^2 + 2 + b + 3)t + (b + 3)(b^2 + 2) = 0$$

$$\Rightarrow t = b^2 + 2, b + 3$$

For real solution

$$0 \leq b^2 + 2 \leq 5 \quad \text{or} \quad 0 \leq b + 3 \leq 5$$

$$-2 \leq b^2 \leq 3 \quad \text{or} \quad -3 \leq b \leq 2$$

12. Ans (A,B,C)

$$(1) D \geq 0$$

$$(2) -6 < \frac{-b}{2a} < 1$$

$$(3) f(-6) > 0 \text{ and } f(1) > 0$$

$$\therefore 6 \leq a < 6.75$$

$$[a] = 6$$

i. e., $2, h_1, h_2, \dots, h_{20}, [a]$ are in HP.

$\Rightarrow 2, h_1, h_2, \dots, h_{20}, 6$ are in H.P.

$$\Rightarrow \frac{1}{h_{18}} = \frac{1}{2} + 18 \left[\frac{\frac{1}{6} - \frac{1}{2}}{21} \right] \Rightarrow h_{18} = \frac{14}{3}$$

$2, a_1, a_2 \dots a_{20}, 6$ are in A.P.

$$\therefore a_3 = 2 + 3 \left(\frac{6 - 2}{21} \right) = \frac{18}{7}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5.00)

Centre of C_3 will lie on the radical axis of C_1 and C_2 which is $10x + 6y + 26 = 0$.

Let center of C_3 is (h, k) .

Equation of chord of contact through (h, k) to the

C_1 may be given as $hx + ky = 25 \dots (I)$

Let the mid point of the chord is (x_1, y_1) the

equation of the chord with the help of mid point may be

$$\text{given as } xx_1 + yy_1 = x_1^2 + y_1^2$$

Since (I) and (II) represents same straight line

$$\Rightarrow \frac{h}{25} = \frac{x_1}{x_1^2 + y_1^2}, \frac{k}{25} = \frac{y_1}{x_1^2 + y_1^2} \dots (II)$$

Since (h, k) lie on the radical axis

$$10 \left(\frac{25x_1}{x_1^2 + y_1^2} \right) + 6 \left(\frac{25y_1}{x_1^2 + y_1^2} \right) + 26 = 0$$

$$\text{the locus of } (x_1, y_1) \text{ is } 5x + 3y + \frac{13}{25}(x^2 + y^2) = 0$$

2. Ans (1.00)

$$f(n, \theta) = \prod_{r=1}^n \left(1 - \frac{\sin^2 \frac{\theta}{2^r}}{\cos^2 \frac{\theta}{2^r}} \right)$$

$$f(n, \theta) = \prod_{r=1}^n \left(\frac{\cos \frac{\theta}{2^{r-1}}}{\cos \frac{\theta}{2^r}} \right) ;$$

$$f(n, \theta) = \frac{\cos \theta \cdot \cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \dots \cos \frac{\theta}{2^{n-1}}}{\left(\cos \frac{\theta}{2} \cdot \cos \frac{\theta}{2^2} \cdot \cos \frac{\theta}{2^3} \dots \dots \cos \frac{\theta}{2^n} \right)^2}$$

$$f(n, \theta) = \frac{\frac{\sin 2^n \frac{\theta}{2^{n-1}}}{2^n \cdot \sin \frac{\theta}{2^{n-1}}}}{\left(\frac{\sin 2^n \cdot \frac{\theta}{2^n}}{2^n \cdot \sin \frac{\theta}{2^n}} \right)^2} ;$$

$$f(n, \theta) = \frac{(\sin 2\theta) \left(2^{2n} \sin^2 \frac{\theta}{2^n} \right)}{(\sin^2 \theta) \left(2^n \sin \frac{\theta}{2^{n-1}} \right)}$$

$$f(n, \theta) = \lim_{n \rightarrow \infty} \frac{\sin 2\theta}{\sin^2 \theta} \cdot \left(\frac{2^{2n} \cdot \sin^2 \frac{\theta}{2^n}}{2^n \cdot \sin \frac{\theta}{2^{n-1}}} \right)$$

$$= \frac{\sin 2\theta}{\sin^2 \theta} \cdot \frac{\theta^2}{2\theta} = \frac{\theta}{\tan \theta}$$

$$g(\theta) = \lim_{\theta \rightarrow 0} g(\theta) = 1$$

3. Ans (0.00)

$$\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} = \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix}$$

$$abc(a-b)(b-c)(c-a)(ab+bc+ca) = (a-b)$$

$$(b-c)(c-a)(a+b+c)$$

$$\therefore abc(ab+bc+ca) = (a+b+c)$$

4. Ans (141.00)

Here the sum of the numbers are vanishes of six cards i.e

Case I : If selected 3 cards each of number -1 or 1 i.e

$$\text{The number of arrangement} = \frac{6!}{3!3!} = 20$$

Case II : If selected 2 cards each of no. -1, 0 or 1 i.e

$$\text{number of arrangement} = \frac{6!}{2!2!2!} = 90$$

Case III : If selected one card each of number -1 and 1 and 4 cards of no. 0.

$$\text{so no. of arrangement is } \frac{6!}{1!1!4!} = 30$$

Case IV : If all cards selected from the no. 0

$$\text{So no. of arrangement is } \frac{6!}{6!} = 1$$

Hence total no. of arrangement is $20 + 90 + 30 + 1 = 141$

5. Ans (1.69)

$$\cos(\alpha + \beta) = \frac{4}{5}, \sin(\alpha - \beta) = \frac{5}{13},$$

$$\left(\begin{array}{l} \because 0 < \alpha, \beta < \pi/4 \\ \Rightarrow 0 < \alpha + \beta < \pi/2 \end{array} \right)$$

$$\Rightarrow \tan 2\alpha = \tan \{(\alpha + \beta) + (\alpha - \beta)\}$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)}$$

$$\text{here } \tan(\alpha + \beta) = 3/4, \tan(\alpha - \beta) = 5/12$$

$$\Rightarrow \tan 2\alpha = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \times \frac{5}{12}} = \frac{56}{48 - 15}$$

$$\Rightarrow \tan 2\alpha = \frac{56}{33}$$

6. Ans (24.00)

$$\frac{2dx}{dy} = 3 \left(\frac{x}{y} \right) - \left(\frac{y}{x} \right), \text{ put } x = vy$$

$$2v + 2y \frac{dv}{dy} = 3v - \frac{1}{v}$$

$$\int \frac{2v}{v^2 - 1} dv = \int \frac{dy}{y}$$

$$\Rightarrow x^2 = y^2 + cy^3$$

$$\Rightarrow c = 1 \text{ (passing through (6, 3))}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C,D	B,C	A,C	A,B,D	A,C	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	20.00	1.58	2.50	0.50	66.00	37.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	7	2	2	3	6

PART-2 : CHEMISTRY

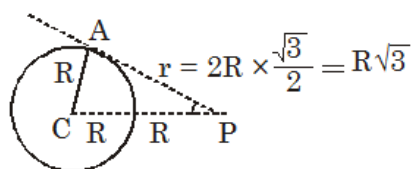
SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	D	A,B,D	A,B,C,D	B,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.40	247.74	36.00	5.00	3.00	3.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	5	3	3	2	5

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,D	A,B,C	A,D	B,C,D	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.07	0.20	216.00	3.00	13.38	0.59
SECTION-III	Q.	1	2	3	4	5	6
	A.	9	6	5	3	4	5

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,C,D)



Potential at center and at any point on the sphere will be same

$$\text{So, } v_A = v_C = \frac{p}{4\pi\epsilon_0 (2R)^2} = \frac{p}{16\pi\epsilon_0 R^2}$$

$$v_A \text{ due to dipole} = \frac{KP \cos \theta}{(\sqrt{3}R)^2}$$

2. Ans (B,C)

$$v_L = v_C = v_R ;$$

$$\Rightarrow x_L = x_C = R$$

when inductor is short circuited

$$Z = \sqrt{R^2 + x_C^2} = \sqrt{2}R$$

$$\therefore I = \frac{30}{Z} = \frac{30}{\sqrt{2}R}$$

$$\therefore V_L = ix_L = \frac{30}{\sqrt{2}R} \times R = \frac{30}{\sqrt{2}}$$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here f_0 is the resonance frequency as $v_L = v_C$

$$\Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ and } \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

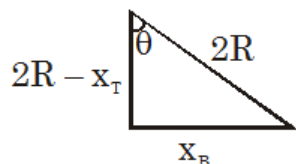
$$\frac{x_L}{x_C} = \frac{\omega L}{\sqrt{\omega C}} = \omega^2 LC$$

$$\text{Given } f = 2f_0$$

$$\Rightarrow \omega = 2\omega_0$$

3. Ans (A,C)

Consider a moment when the center of top cylinder has moved down a distance x_T and the center of the bottom cylinder has moved to the right a distance x_B . At this moment, the cylinders are moving with speeds v_T and v_B , respectively. From figure, v_T and v_B , respectively.



From figure.

$$x_T = 2R(1 - \cos \theta)$$

Also energy conservation gives :

$$m \left(\frac{v_B^2 + v_T^2}{2} \right) = mgx_T = 2mgR(1 - \cos \theta)$$

$$x_B^2 + (2R - x_T)^2 = (2R)^2$$

Differentiating w.r.t time

$$v_T = v_B x_B / (2R - x_T) = v_B \tan \theta.$$

Eliminating v_T from the energy conservation equation gives $v_B^2 = \frac{4gR(1 - \cos \theta)}{(1 + \tan^2 \theta)}$

Setting the derivative with respect to θ equal to zero gives $\cos \theta_m = 2/3$

for the angle θ_m at which v_B is maximum. At this point, the speed of the bottom cylinder is

$$v_B = \sqrt{\frac{16gR}{27}}.$$

Mathematically, it appears that the speed decreases after θ_m . However, for this to happen, the acceleration and, therefore, the horizontal force must be directed to the left (negative) for angles greater than θ_m . But the contact force on the bottom cylinder by the top cylinder cannot be directed to the left. Therefore, the bottom cylinder loses contact with the top and moves off at the maximum speed of $v_B = \sqrt{\frac{16gR}{27}}$

4. Ans (A,B,D)

Perimeter is decreasing at a rate of $2v$

$$\therefore \frac{d}{dt}(2\pi r) = 2v \Rightarrow \frac{dr}{dt} = \frac{v}{\pi}$$

$$\therefore r = \left(r_0 - \frac{v}{\pi} t \right)$$

$$\phi = B \cdot \pi r^2 \Rightarrow \varepsilon = \left| \frac{-d\phi}{dt} \right| = B \cdot 2\pi \cdot r \frac{dr}{dt}$$

$$\therefore \varepsilon = 2B\pi \left(r_0 - \frac{v}{\pi} t \right) \cdot \frac{v}{\pi} = 2Bv \left(r_0 - \frac{v}{\pi} t \right)$$

$$I = \frac{\phi}{R} = \frac{2Bvr}{\lambda \cdot 2\pi r} = \frac{Bv}{\pi \lambda}$$

6. Ans (B,C)

Difference in path travelled by the sound = hd/D

Hence time gap between the recordings = hd/Dv

For resultant pressure amplitude to be maximum, path difference. $\Delta D = m\lambda$

$$\frac{hd}{D} = m \frac{v}{f}$$

$$\left[f = \frac{mvD}{hd} \right]$$

For minimum frequency, $m = 1$

$$\left[f = \frac{vD}{hd} \right]$$

PART-1 : PHYSICS

SECTION-II

1. Ans (20.00)

Just after the thread is burnt

Spring force = $600 \times 0.01 = 6\text{N}$ remains same

Torque equation = $I\alpha = 6 \times 0.60 + 20 \times 0.10$ (c.w.)

$$\Rightarrow 2 \left(\frac{1}{12} + 0.01 \right) \alpha = 5.6$$

$$\Rightarrow \alpha = \frac{5.6 \times 12}{2 \times 1.12} = 30$$

$$\text{Applying N2L} = 6 + 20 - N = 2 \times 0.10 \times \frac{5.6 \times 12}{2 \times 1.12}$$

$$\Rightarrow N = 20$$

2. Ans (1.58)

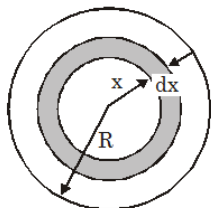
N2L = block

$$T - 50 \times 10 = 50 \times 2 \Rightarrow T = 600 \text{ N}$$

$$\& 1 \times \frac{2}{0.25} = F \times 0.10 - 0.25 \times 600$$

$$\Rightarrow 0.10 F = 8 + 150$$

3. Ans (2.50)



$$d\phi = (2\pi x dx) kx^2$$

$$\phi = \frac{2}{3} kx^2 \pi x^3$$

$$\frac{d\phi}{dt} = \frac{4\pi kx^3}{3}$$

$$E 2\pi x = \frac{4\pi kx^3}{3};$$

$$E = \frac{2}{3} kx^2;$$

$$d\tau = \left(\frac{2}{3} kx^2 \right) \frac{Q}{\pi R^2} (2\pi x dx) x$$

$$\int d\tau = \frac{4}{3} \frac{kQ}{R^2} \int_0^R x^4 dx \Rightarrow \tau = \frac{4}{3} \frac{kQ}{R^2} \frac{R^5}{5}$$

$$\text{At } t = 15 \text{ sec, } \tau = 2.50 \text{ Nm}$$

4. Ans (0.50)

$$\frac{\vec{V} \cdot (\vec{V} - \vec{U})}{\vec{U} \cdot (\vec{V} - \vec{U})} = e$$

5. Ans (66.00)

$$\frac{1}{f_L} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{20} + \frac{1}{20} \right)$$

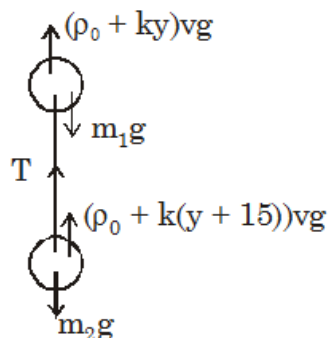
$$\Rightarrow f_L = 20 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{-10} - \frac{2}{20} = -\frac{1}{5}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-5} \Rightarrow \frac{1}{v} = \frac{1}{30} - \frac{1}{5}$$

$$\text{For lens } \frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

6. Ans (37.50)



$$1 + 0.01 y + 1 + 0.01 y + 0.15 = 2.6$$

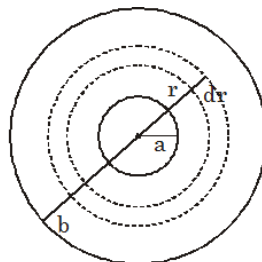
$$0.02y = 0.45$$

PART-1 : PHYSICS

SECTION-III

2. Ans (7)

At any distance r from the axis



$$B = \frac{\mu_0}{2\pi r}$$

$$\therefore \text{Field energy density} = \frac{B^2}{2\mu_0} = \frac{\mu_0 i^2}{8\pi^2 r^2}$$

[Magnetic energy density]

$$\int du_B = \int_a^b u_B dv$$

$$\Rightarrow \frac{\mu_0 i^2}{8\pi^2} \left(\frac{2\pi r dr \ell}{r^2} \right) \Rightarrow \frac{\mu_0 i^2 \ell}{4\pi} \int_a^b \frac{dr}{r} \Rightarrow \frac{\mu_0 i^2 \ell}{2\pi} \ln 2$$

3. Ans (2)

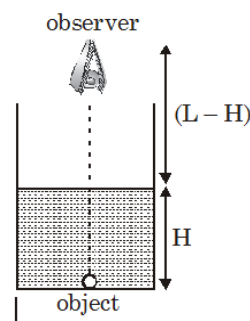
$$\frac{dN_A}{dt} = -\lambda_1 N_A, \frac{dN_B}{dt} = 2\lambda_1 N_A - \lambda_2 N_B$$

$$N_B = \text{maximum} \Rightarrow \frac{dN_B}{dt} = 0$$

$$\Rightarrow 2\lambda_1 N_A = \lambda_2 N_{B_{\max}} \Rightarrow N_{B_{\max}} = \frac{2\lambda_1}{\lambda_2} N_A$$

$$\Rightarrow N_{B_{\max}} = \frac{2\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t} = 2$$

4. Ans (2)



$$X_{\text{app}} = L - H + \frac{H}{\mu}$$

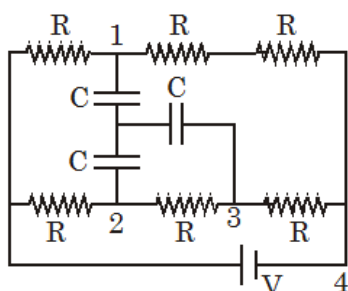
$$\frac{dX_{\text{app}}}{dt} = \frac{dH}{dt} \left(\frac{1-\mu}{\mu} \right)$$

$$\pi r^2 H = V$$

$$\therefore \frac{dH}{dt} = -\frac{2H}{r} \frac{dr}{dt} = \frac{2KH}{r}$$

$$\therefore \frac{dX_{\text{app}}}{dt} = \frac{2KV}{\pi r^3} \left(\left(\frac{1-\mu}{\mu} \right) \right)$$

6. Ans (6)



No current passes through capacitors in steady state. Assume potential at point '4' to be zero.

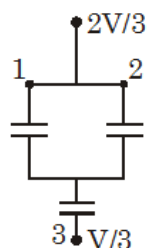
Then points '1' and '2' are at same potential $\frac{2V}{3}$

Hence C_1 and C_2 can be taken in parallel

The potential at point 3 is $\frac{V}{3}$

$\therefore$ Equivalent circuit of all three capacitors is shown Hence potential difference across capacitor C_3 is

$$= \frac{2C}{2C + C} \times \left(\frac{2V}{3} - \frac{V}{3} \right) = \frac{2V}{9}$$



PART-2 : CHEMISTRY

SECTION-I

1. Ans (B,C)

$$\eta_{H_2} = \frac{1 \times 0.1}{0.08 \times 300} = 4.166 \times 10^{-3}$$

92 mole $H_2 \rightarrow 1$ mole Ru

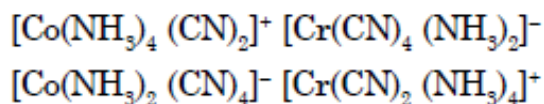
$$4.166 \times 10^{-3} \rightarrow \frac{1}{92} \times 4.1666 \times 10^{-3} \text{ mol Ru}$$

$$= \left(\frac{4.166}{92} \times 101 \right) \text{ mg Ru}$$

$$\text{Moles of } NaBH_4 = \frac{\eta_{H_2}}{4} = \frac{4.166 \times 10^{-3}}{4} = 1.0415 \times 10^{-3}$$

$$\text{time} = \frac{0.1 \times 1}{1.0415 \times 10^{-3}} = 96 \text{ min.}$$

4. Ans (A,B,C,D)



6. Ans (B,C,D)

This is the test for 'N' & known as 'Laigne Test'. In which, on reaction with Na, 'N-containing' compound form cyanide ion; and give Prussian blue color with $FeSO_4$.

*[Hydrazine, Hydroxylamine & NH_3 do not form Cyanide ion; because 'C' is absent; Hence they do not give Prussian blue color with $FeSO_4$]

(A) Carbon is absent; Cyanide ion formation does not take place.

$H_2N - NH_3^+ Cl^-$ (They do not give Prussian blue color)

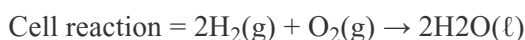
(B), (C), (D) $\rightarrow$ They all give Prussian blue color with $FeSO_4$.

Ans. (B), (C), (D)

PART-2 : CHEMISTRY

SECTION-II

1. Ans (0.40)

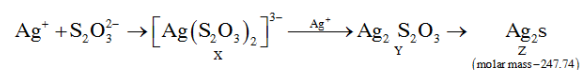


$$\Delta_r G^\circ = 2(-236 \times 1000) = -4 \times \frac{236000}{2.46} \times E_{\text{cell}}^\circ$$

$$E_{\text{cell}}^\circ = 1.23$$

$$E_{\text{cathode}}^\circ + 0.83 = 1.23 ; E_{\text{cathode}}^\circ = 0.40 \text{ V}$$

2. Ans (247.74)



3. Ans (36.00)

$$E = q \cdot V. \Rightarrow 2eV = 2 \times 1V$$

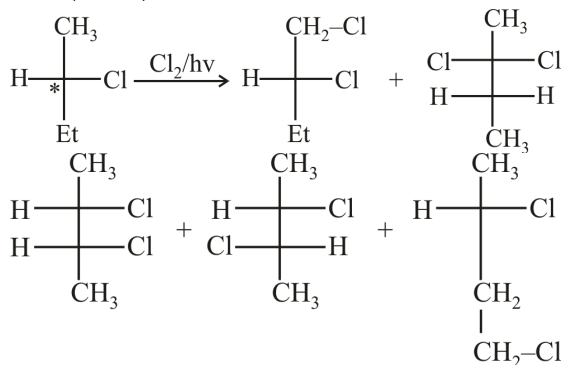
$\therefore$ charge on ion = -2 (as it moves towards positive plate)

Let the ion is ${}^Z_79X^{2-}$. The number of electron is (Z + 2) and number of neutrons is (79 - Z).

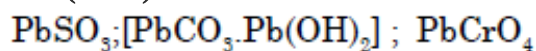
$$\text{From question, } (Z + 2) \times 1.25 = 79 - Z \Rightarrow Z = 34$$

$$\therefore \text{Number of electrons} = Z + 2 = 36$$

4. Ans (5.00)



6. Ans (3.00)



PART-2 : CHEMISTRY

SECTION-III

1. Ans (2)

$$T_C = \sqrt{\frac{8\alpha}{27R\beta}}$$

2. Ans (5)

$$\Delta T_f = K_f \times m \times i$$

$$7 = 14 \times \frac{75.2}{94} \times \left(1 - \alpha + \frac{\alpha}{2}\right)$$

$$\alpha = 0.75 = 75\%$$

5. Ans (2)

(i) & (iii)

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B,C)

$$g(x) = 2f\left(\frac{x}{2}\right) + f(1-x) \text{ and } g'(x) = f'(x/2) - f'(1-x)$$

Now $g(x)$ is increasing if $g'(x) \geq 0$

$$f'\left(\frac{x}{2}\right) \geq f'(1-x)$$

$[\because f'(x) < 0 \text{ i.e. } f(x) \text{ is decreasing}]$

$$\Rightarrow \frac{x}{2} \leq 1-x \Rightarrow x \leq 2-2x$$

$$\Rightarrow 3x \leq 2 \Rightarrow x \leq 2/3 \Rightarrow 0 \leq x \leq \frac{2}{3}$$

$\Rightarrow g(x)$ increases in $0 \leq x \leq 2/3$

and $g'(x) < 0$ for decreasing

$$\Rightarrow f'\left(\frac{x}{2}\right) \leq f'(1-x) \Rightarrow \frac{x}{2} \geq 1-x$$

$$\Rightarrow x \geq 2/3 \leq 2/3 \leq x \leq 1$$

2. Ans (A,B,D)

Given that z and w are two complex numbers. To prove

$$|z|^2 w - |w|^2 z = z - w \Leftrightarrow z = w \text{ or } z\bar{w} = 1$$

First let us consider

$$|z|^2 w - |w|^2 z = z - w \quad \dots(1)$$

$$\Rightarrow z(1 + |w|^2) = w(1 + |z|^2)$$

$$\Rightarrow \frac{z}{w} = \frac{1 + |z|^2}{1 + |w|^2}$$

$$\Rightarrow \overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}} \Rightarrow \frac{\bar{z}}{\bar{w}} = \frac{z}{w}$$

$$\Rightarrow \bar{z}w = z\bar{w} \quad \dots(2)$$

Again from Eq. (1),

$$z\bar{z}w - w\bar{w}z = z - w$$

$$z(\bar{z}w - 1) - w(\bar{w}z - 1) = 0$$

$$z(z\bar{w} - 1) - w(z\bar{w} - 1) = 0 \quad [\text{Using Eq. (2)}]$$

$$\Rightarrow (z\bar{w} - 1)(z - w) = 0$$

$$\Rightarrow z\bar{w} = 1 \text{ or } z = w$$

3. Ans (A,B,C)

$$(i) \quad z_1 \bar{z}_1 = 1 \Rightarrow \bar{z}_1 = \frac{1}{z_1}$$

$$(ii) \quad |z_1 + z_2 + \dots + z_n| = |\bar{z}_1 + \bar{z}_2 + \dots + \bar{z}_n|$$

$$= \left| \frac{1}{z_1} + \frac{1}{z_2} + \dots + \frac{1}{z_n} \right|$$

$$z_1 + \frac{1}{z_1} = z_1 + \bar{z}_1 = 2 \operatorname{Re}(z_1)$$

$\therefore$ Centroid =

$$\frac{2 [\operatorname{Re}(z_1) + \operatorname{Re}(z_2) + \dots + \operatorname{Re}(z_n)]}{n}$$

Whose imaginary part = 0

4. Ans (A,D)

$$f(x) = \int_0^x f(t) \cos t - \int_0^x \cos(t-x) dt$$

$$f(x) = \int_0^x f(t) \cos t - \int_0^x \cos t dt$$

$$f'(x) = f(x)\cos x - \cos x$$

$$\Rightarrow \frac{dy}{dx} = y\cos x - \cos x \quad (y = f(x))$$

$$\Rightarrow \frac{dy}{dx} - y\cos x = -\cos x$$

$$\text{I.F.} = e^{\int -\cos x dx} = e^{-\sin x}$$

$$\Rightarrow y \cdot e^{-\sin x} = \int -e^{-\sin x} \cdot \cos x dx$$

$$y \cdot e^{-\sin x} = e^{-\sin x} + c$$

$$\Rightarrow y = ce^{\sin x} + 1$$

$$y = 0 \text{ when } x = 0$$

$$\Rightarrow c = -1$$

$$\Rightarrow f(x) = 1 - e^{\sin x}$$

$$f(0) = f(0) - 1 = -1$$

$$f'(x) = f(x)(-\sin x) + \cos \cdot f(x) + \sin x$$

$$f''\left(\frac{\pi}{2}\right) = -f\left(\frac{\pi}{2}\right) + 1$$

$$= -1 + e + 1 = e$$

5. Ans (B,C,D)

Since $|\sin \theta|$ is continuous $f(x)$ will be differentiable

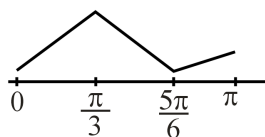
$$f'(x) = \left| \sin x \left(x + \frac{\pi}{3} \right) - |\sin x| \right|$$

$$f(x) = 0 \Rightarrow \sin^2 \left(x + \frac{\pi}{3} \right) = \sin^2 x$$

$$\Rightarrow \sin \left(2x + \frac{\pi}{3} \right) \cdot \sin \frac{\pi}{3} = 0$$

$$\Rightarrow 2x + \frac{\pi}{3} = \pi, 2\pi \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{6}$$

Indicator diagram of $f(x)$ is



$$f(0) = \frac{1}{2}$$

$$f\left(\frac{\pi}{3}\right) = 1$$

$$f\left(\frac{5\pi}{6}\right) = 2 \int_{5\pi/6}^{\pi} \sin x dx = 2 - \sqrt{3}$$

$$f(\pi) = \frac{1}{2}$$

6. Ans (A,D)

$$I = \int \frac{3 \cot 3x - \cot x}{\tan x - 3 \tan 3x} dx = pf(x) + g(x) + C \quad \dots (i)$$

$$\text{using } \tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x} \quad \text{and}$$

$$\cot 3x = \frac{\cot^3 x - 3 \cot x}{3 \cot^2 x - 1}$$

$$I = \int \frac{1 - 3 \tan^2 x + 2 - 2}{3 - \tan^2 x} dx$$

$$I = \int 1 \cdot dx - 2 \int \frac{1 + \tan^2 x}{3 - \tan^2 x} dx = x - 2 \int \frac{\sec^2 x}{3 - \tan^2 x} dx + C$$

$$= x - \frac{2}{2\sqrt{3}} \ln \left| \frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x} \right| + C$$

$$= 1 \cdot x + \frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x} \right| + C$$

comparing with R.H.S. of equation (i), we get the result

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.07)

$$\text{Let } t^3 = x$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin x - x \cos x + x^5}{x(e^{3x} - 1 - 3x)}$$

Now use expansions of $\sin x$, $\cos x$ and e^{3x}

2. Ans (0.20)

$$f(x) = k(x-1)(x-3)(x-2)^2 \text{ and } k > 0$$

$$\Rightarrow f(x) = k \int (x^2 - 4x + 3)(x^2 - 4x + 4) dx$$

$$= k \int x^4 + 16x^2 - 8x^3 + 7(x^2 - 4x) + 12 dx$$

$$= k \left(\frac{x^5}{5} - 2x^4 + \frac{23x^3}{3} - 14x^2 + 12x \right) + c$$

$$\therefore f(0) = 2 \Rightarrow c = 2$$

$$\therefore f(1) = \frac{88}{15} \Rightarrow$$

$$\frac{88}{15} = k \left(\frac{1}{5} - 2 + \frac{33}{3} - 14 + 12 \right) + c$$

$$\Rightarrow \frac{58}{15} = k \times \frac{58}{15} \Rightarrow k = 1 \therefore a_5 = \frac{1}{5}$$

3. Ans (216.00)

Probability of defective screw $p = \frac{1}{10}$ and

non-defective screw $q = \frac{9}{10}$

Probability that a pocket of 10 screws would have to be replaced = probability that pocket contains three or more defective screw

$$p(x) = 1 - p(x < 3) = 1 - \{p(x=0) + p(x=1) + p(x=2)\}$$

$$= 1 - \left\{ {}^{10}C_0 \left(\frac{1}{10}\right)^0 \left(\frac{9}{10}\right)^{10} + {}^{10}C_1 \left(\frac{1}{10}\right)^1 \left(\frac{9}{10}\right)^9 + {}^{10}C_2 \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^8 \right\}$$

$$= 1 - \left\{ \frac{9^{10} + 10(9^9) + 45(9^8)}{10^{10}} \right\}$$

$$= 1 - \frac{9^8(81 + 90 + 45)}{10^{10}} = 1 - \frac{9^8(216)}{10^{10}}$$

Hence, $\lambda = 216$

4. Ans (3.00)

$$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$$

$$= 2(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$= 52 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \leq 78$$

$$\text{Because, } |\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0$$

$$\Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq -26$$

Hence, $12k + \lambda = 78$ where $k = 2, \lambda = 54$

$k = 3; \lambda = 42$ and $k = 5; \lambda = 18$

Thus, three values

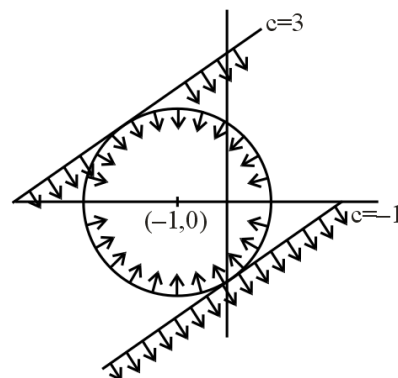
5. Ans (13.38)

$$\left| \frac{-5 - 0 + c}{13} \right| = \sqrt{2}$$

$$\Rightarrow c - 5 = \pm 13\sqrt{2}$$

$$\Rightarrow c = 5 \pm 13\sqrt{2}$$

but $c \leq 0$ hence $c = 5 - 13\sqrt{2}$



But $c = 5 - 13\sqrt{2}$ common point is one

$c = 5 + 13\sqrt{2}$ common point is infinite

Hence $c = 5 - 13\sqrt{2}$ is Answer.

6. Ans (0.59)

$$I_1 = \int_0^{\pi/2} (\cos x)^{\sqrt{2}} \cos x \, dx$$

$$= (\cos x)^{\sqrt{2}} \sin x \Big|_0^{\pi/2} + \sqrt{2} \int_0^{\pi/2} (\cos x)^{\sqrt{2}-1} \sin^2 x \, dx$$

$$\Rightarrow I_1 = \sqrt{2}I_2 - \sqrt{2}I_1$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{\sqrt{2}}{1 + \sqrt{2}}$$

$$= 2 - \sqrt{2} = 2 - 1.41 = 0.59$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (9)

$$\lim_{x \rightarrow \infty} \left(x \sin \left(\frac{1}{x} \right) + \sin \left(\frac{1}{x^2} \right) \right) \lim_{x \rightarrow \infty} \left(\frac{\sin \left(\frac{1}{x} \right)}{\frac{1}{x}} + \sin \left(\frac{1}{x^2} \right) \right) = 1 + 0 = 1$$

2. Ans (6)

$$\text{Given } I = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \Rightarrow abc = 120 = 2^3 \times 3^1 \times 5^1$$

Case-I All are positive = ${}^5C_2 \times {}^3C_2 \times {}^3C_2 = 90$

Case-II one positive and two negative

$$= 3 \times ({}^5C_2 \times {}^3C_2 \times {}^3C_2) = 270$$

So number of possible matrices = $90 + 270 = 360$

3. Ans (5)

$$\cot^{-1} \frac{n}{\pi} > \frac{\pi}{6} \Rightarrow -\infty < \frac{n}{\pi} < \cot \frac{\pi}{6}$$

$$n < \pi\sqrt{3} \quad n \in \mathbb{N}, n_{\max} = 5$$

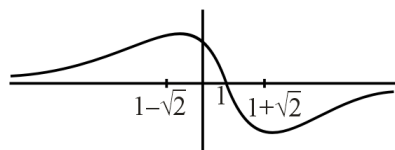
4. Ans (3)

$$f(x) = \frac{1-x}{1+x^2} \Rightarrow f(x) = 0 \text{ at } x = 1 \pm \sqrt{2}$$

for $x \in [-\sqrt{2} + 1, +\sqrt{2}]$ f is bijective function hence f is invertible.

$$\frac{1-x}{1+x^2} = y$$

$$\text{or } x^2y + x + (y-1) = 0$$



or

$$x = \frac{-1 \pm \sqrt{1-4y(y-1)}}{2y} = \frac{-1 \pm \sqrt{4y-4y^2+1}}{2y}$$

$$f^{-1}(x) = \begin{cases} \frac{-1 + \sqrt{4x-4x^2+1}}{2x} & ; x \neq 0 \\ 1 & ; x = 0 \end{cases}$$

$$\text{as } f(1) = 0$$

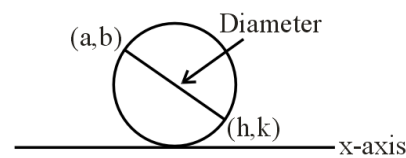
5. Ans (4)

Equation of circle whose diameter's end points are (a, b) and (h, k)

$$(x-a)(x-h) + (y-b)(y-k) = 0$$

$$x^2 + y^2 - x(a+h) - y(b+k) + ah + bk = 0$$

it touches x-axis



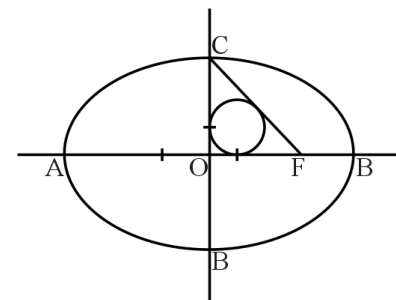
$$\text{Hence } g^2 = c \Rightarrow \left(\frac{a+h}{2} \right)^2 = ah + bk$$

$$\Rightarrow (h-a)^2 = 4bk$$

$$\therefore \text{Locus of } (h, k) \text{ is } (x-a)^2 = 4by$$

6. Ans (5)

$$ae = 6 \Rightarrow b^2 + 36 = (b+4)^2 \Rightarrow 36 = 16 + 8b$$



$$b = \frac{5}{2} \Rightarrow a^2 = a^2e^2 + b^2 = 36 + \frac{25}{4} = \frac{169}{4}$$

$$a = \frac{13}{2} \Rightarrow (2a)(2b) = 65$$